



installations, it is recommended to install the units on rooftops for best results.

Then, high humidity and low temperature reduce the impact of Pure Skies mildly ($\sim 10\%$ at $RH > 70\%$). The company overcame this issue by varying the algorithms to adapt to such adverse conditions.

Dr Sola adds, "With one unit at a coal-fired power plant, pollution levels were down by over 40 per cent, within three to four days. Average baseline level of $PM_{2.5}$ was $61\mu g/m^3$ and PM_{10}

was $117\mu g/m^3$. This reduced to $28\mu g/m^3$ and $69\mu g/m^3$, respectively.

"Another Pure Skies unit was installed at a prime polluted location in New Delhi in the period November-December, 2017. Initial baseline level of $PM_{2.5}$ - 168 and PM_{10} - 285 came down to $PM_{2.5}$ - 88 and PM_{10} - 176, respectively—a reduction of 47 and 38 per cent, respectively—within a couple of days. This data has been acquired from CPCB reference monitor located about seven kilometres away."

The company received initial research support from Ministry of Environment, Forestry and Climate Change. Post successful results at different venues across India, it received interest from various city administrators across the north and north-eastern stretch that have been facing severe levels of pollution. It is now working with several city governments to install a network of Pure Skies in phases across the cities to ensure clean air for all.

—Deepshikha Shukla, technology journalist, EFY

4 AI-POWERED CAMERA That Can Be Your Video Director

For filmmakers and video creators the biggest satisfaction lies in capturing innovative videos from difficult angles smoothly. Manual shooting comes with many limitations that can make taking certain shots hard. As a solution to this, Hong Kong-based Remo Technologies have come up with an AI-powered auto-director camera, called Obsbot Tail. The device comes as a camera lens held by a three-axis gimbal, resting on top of the base body. It has a rotation capacity of 360 degrees. Five TFLOPS of computing power enables it to deliver modern videography features like 3DNR and 3DLUT—it also promises HDR10 capability in the future. All videos can be captured in up to 4K resolution at 60fps, ensuring each motion detail to be captured precisely.

Intelligence of the

camera is powered by HiSilicon Hi3559A chip, which helps it develop features like image-based object-locking and following, auto-zoom, image conditioning and so on. All features are powered by deep learning algorithms. Automated object-locking

makes Obsbot record every move of a specific target, no matter how fast it is moving. The target can be locked using single-touch Tap Lock feature. In fast-paced or complex environments, if the camera misses the target, it can relock the subject with Shar-Lock feature. Pinpointing feature enables the lens to adjust according to distance, position and motion of the subject.

Another interesting feature of Obsbot Tail is its gesture-based control. Powered by sensors and algorithms, it can identify specific gestures of the user or object in front, and perform operations corresponding to the gestures. Operations like start and stop, record, pause, select target, cancel target, zoom and many more can be carried out using Obsbot Tail. Gestures can be customised, too. The feature is supported when the subject or object is standing

